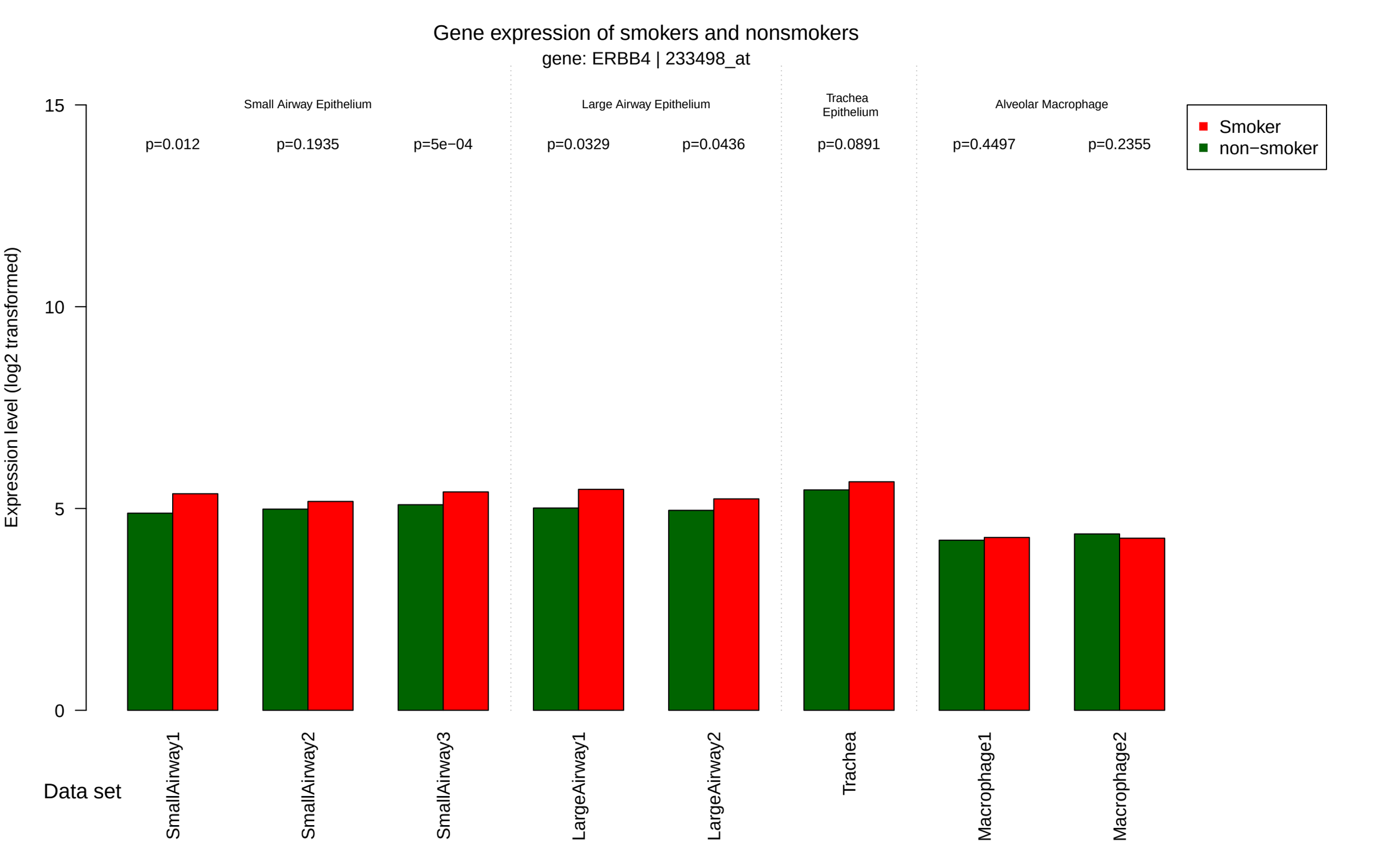
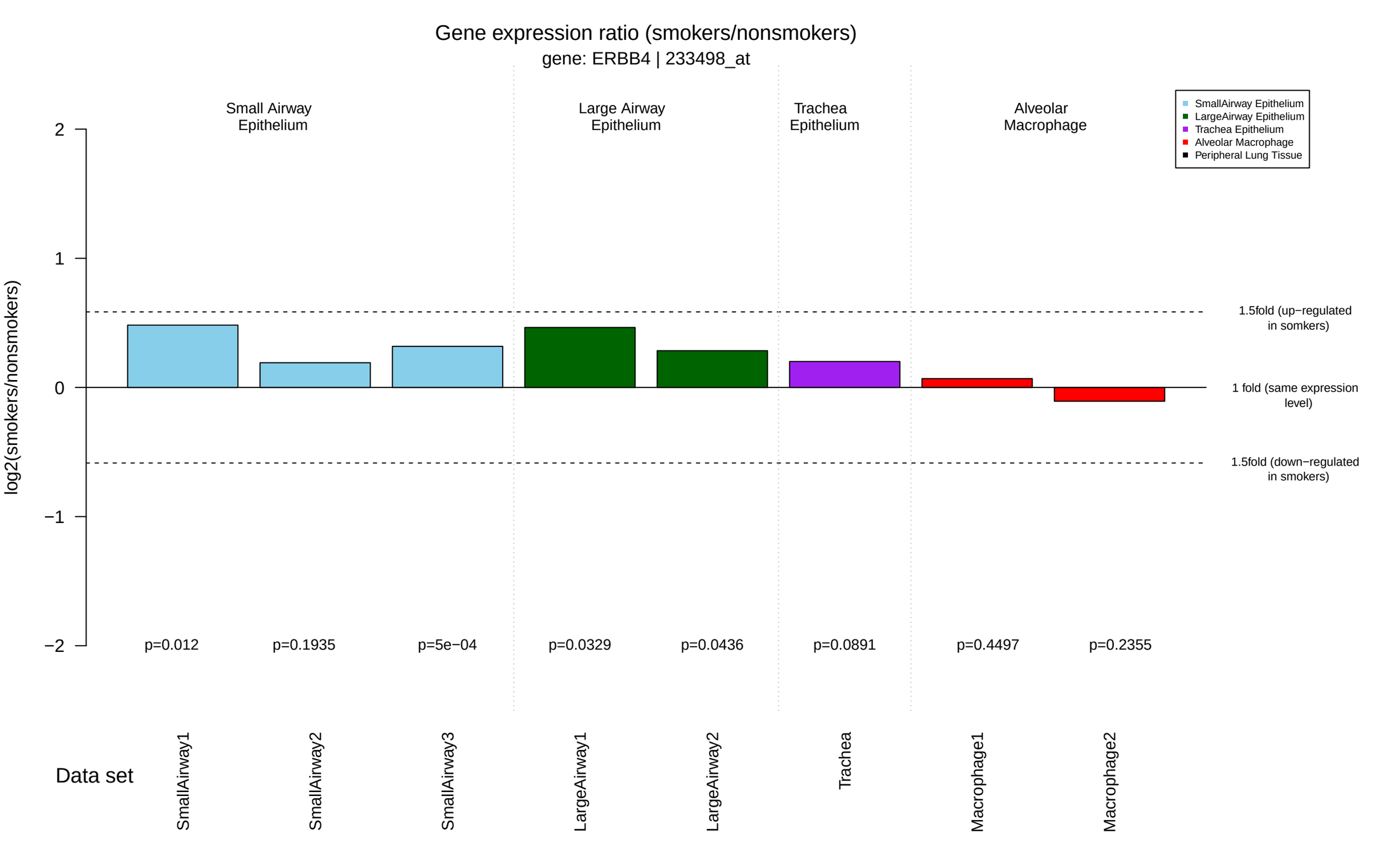
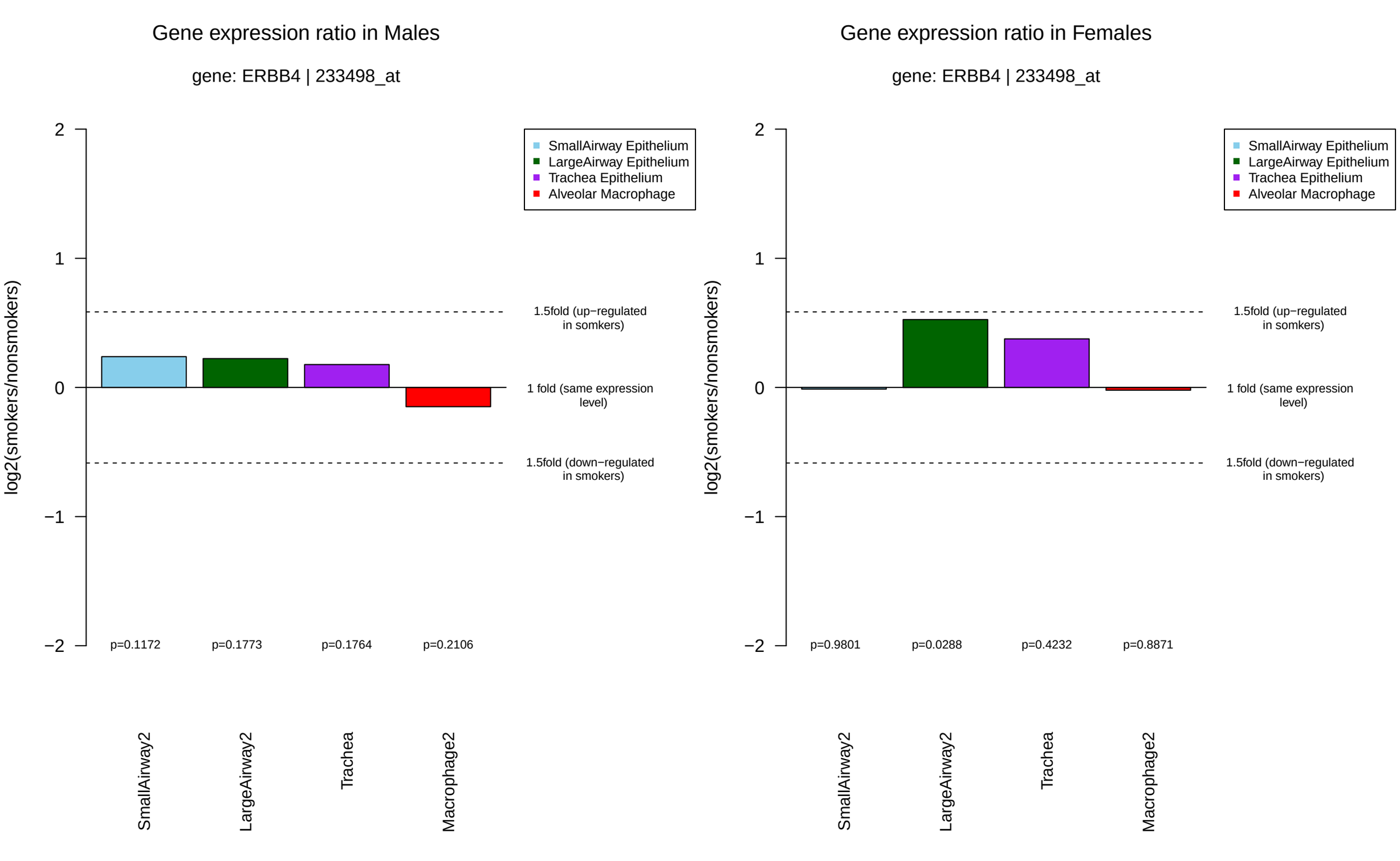
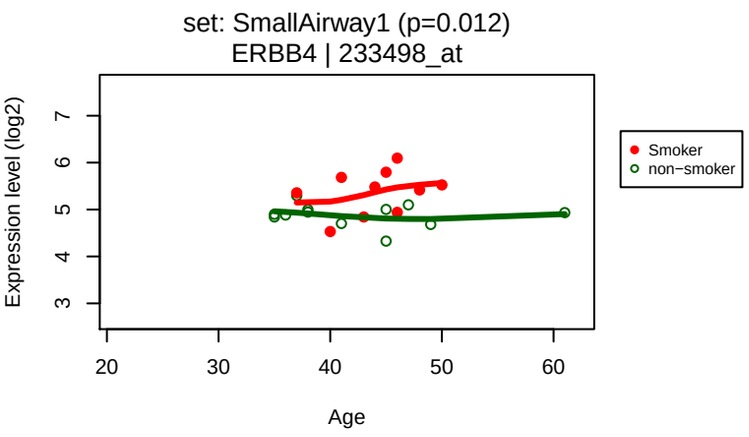
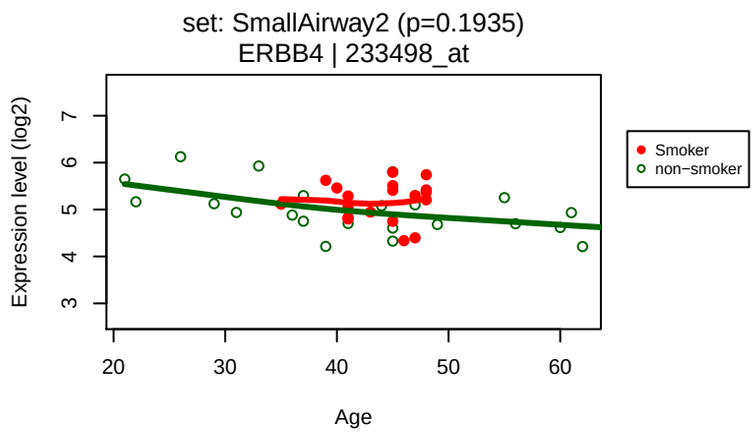
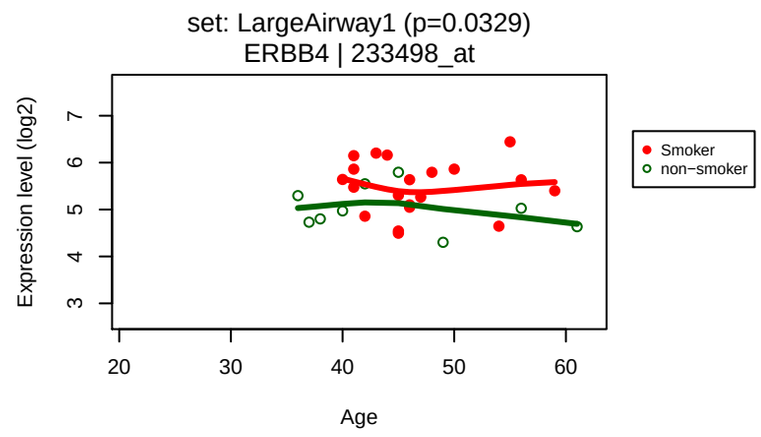

Gene expression of smokers and nonsmokers
gene: ERBB4 | 233498_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.012
p=0.1935
p=5e−04
p=0.0329
p=0.0436
p=0.0891
p=0.4497
p=0.2355
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

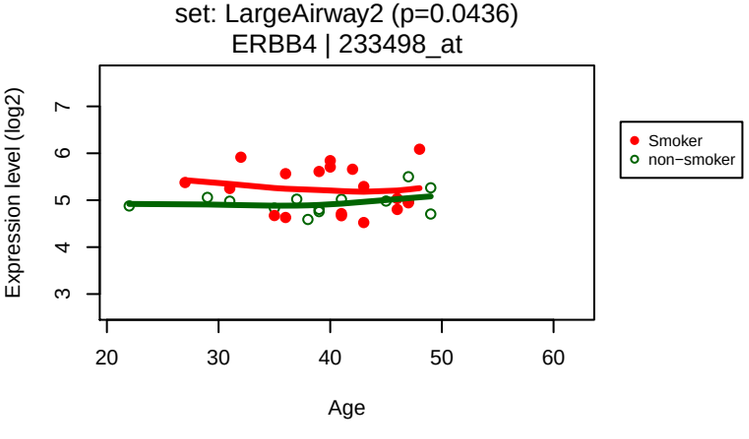

Gene expression ratio (smokers/nonsmokers)
gene: ERBB4 | 233498_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.012
p=0.1935
p=5e−04
p=0.0329
p=0.0436
p=0.0891
p=0.4497
p=0.2355
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

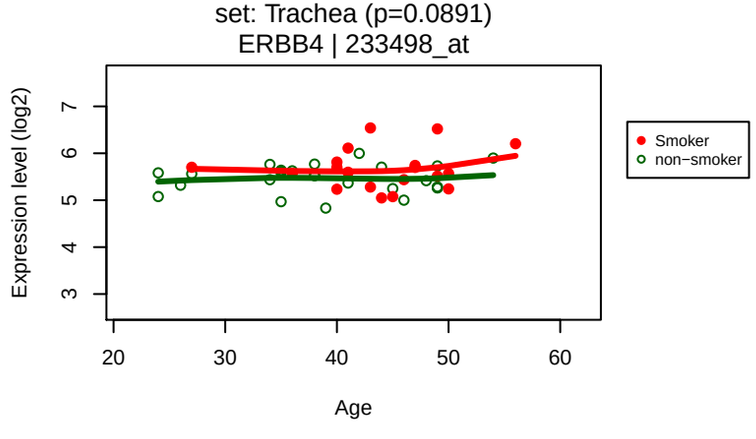

Gene expression ratio in Males
gene: ERBB4 | 233498_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1172
p=0.1773
p=0.1764
p=0.2106
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ERBB4 | 233498_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9801
p=0.0288
p=0.4232
p=0.8871
SmallAirway2
LargeAirway2
Trachea
Macrophage2

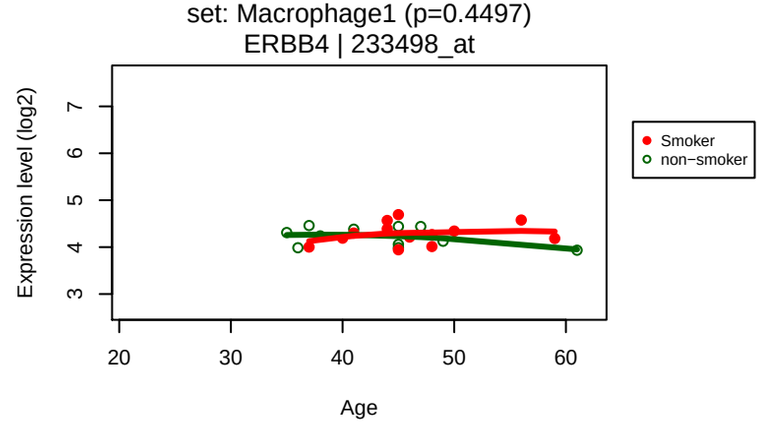

set: SmallAirway1 (p=0.012)
ERBB4 | 233498_at
Expression level (log2)
Age
Smoker
non−smoker

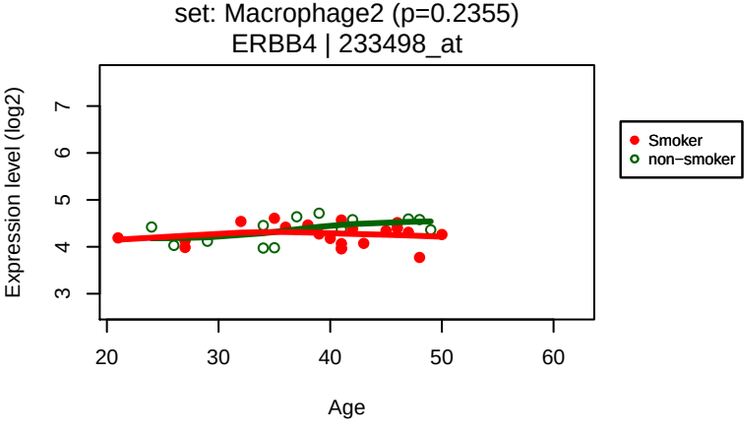

set: SmallAirway2 (p=0.1935)
ERBB4 | 233498_at
Expression level (log2)
Age
Smoker
non−smoker

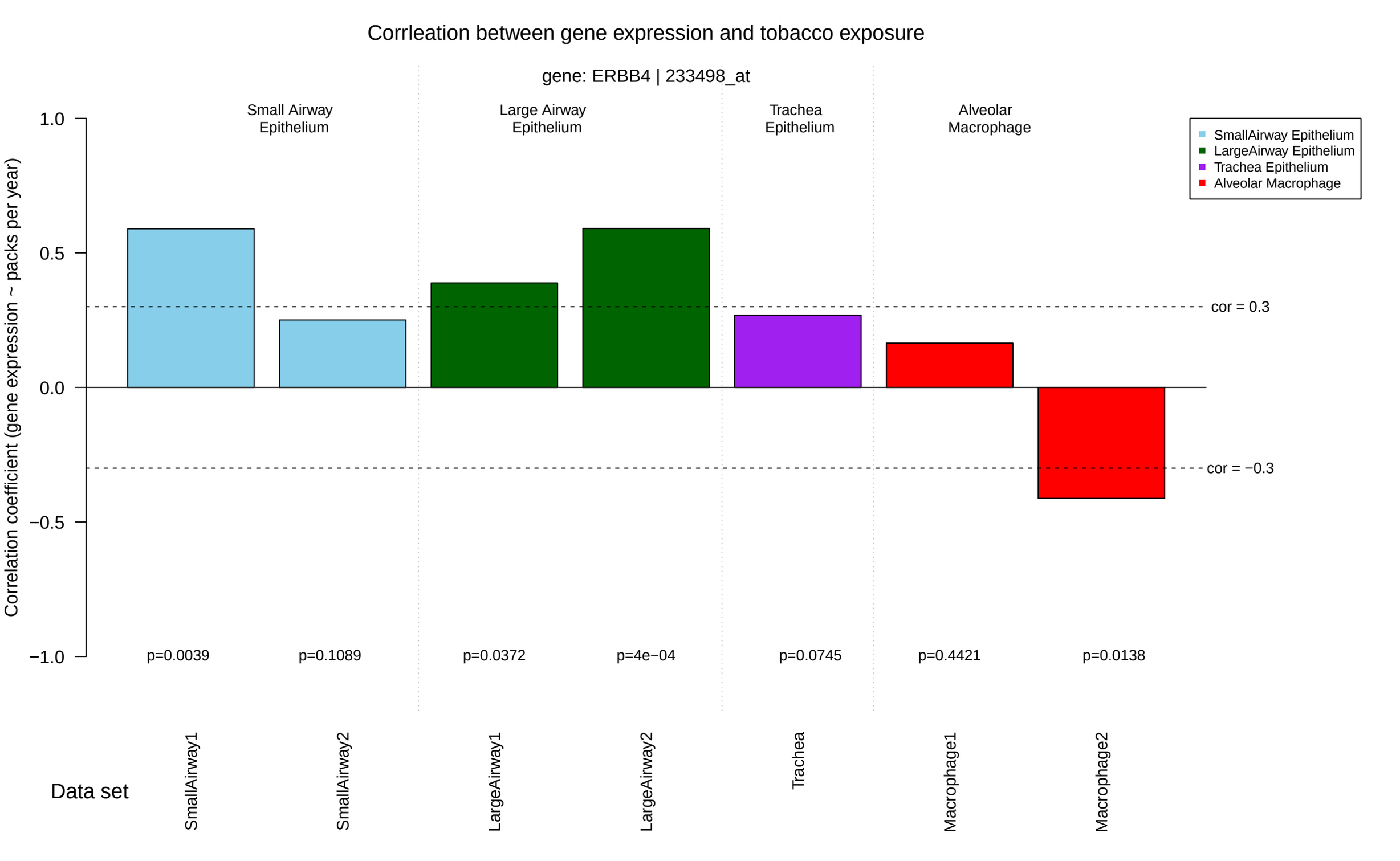

set: LargeAirway1 (p=0.0329)
ERBB4 | 233498_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0436)
ERBB4 | 233498_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0891)
ERBB4 | 233498_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4497)
ERBB4 | 233498_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2355)
ERBB4 | 233498_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ERBB4 | 233498_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0039
p=0.1089
p=0.0372
p=4e−04
p=0.0745
p=0.4421
p=0.0138
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2